

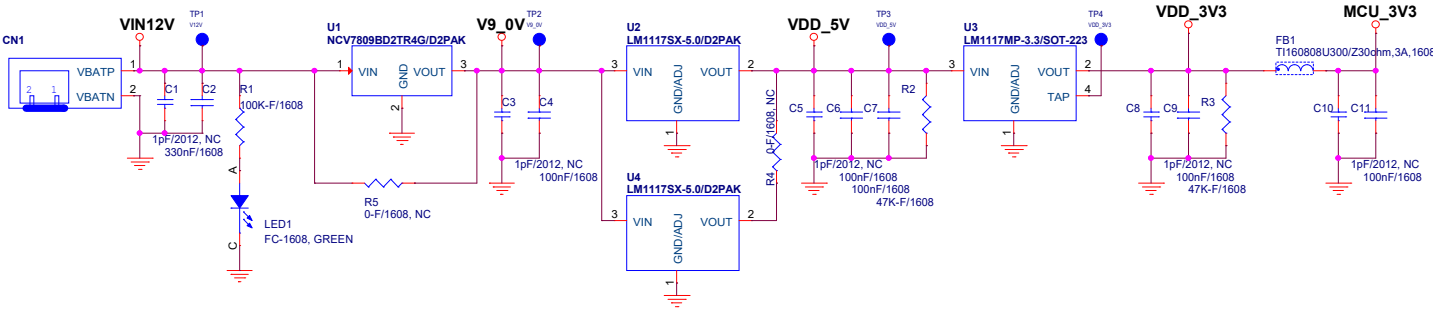
< Rev History >

- 1) '24. 7. 2 - 초안 작성
- 2) '26. 1.30 - 1차 수정
- 3) '26. 5.28 - 양산대응 Test mode 추가

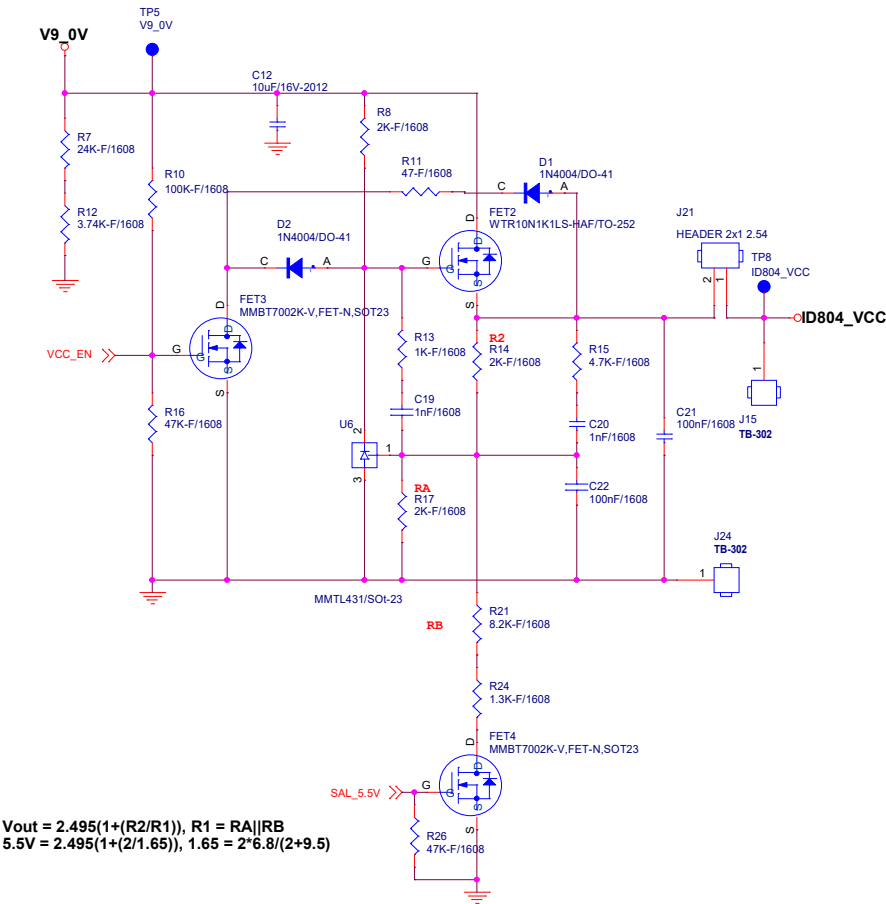
GT Global Technologies 주식회사 글로벌테크놀로지			Design	Check	
Size	Project name ID804 TEST BD		Rev	P.C.G	-
A3	Page name P0. REV HISTORY		2.0		
	Sheet 1 of 5				
	Date: Monday, June 29, 2026				

LOGIC POWER STANDBY

YW396-02V/1R*2-2pin,3.96mm,ST,#16-24,7.96A,DIP,Mate YH396-02V

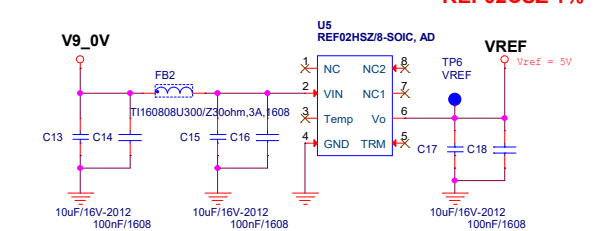


ID804 VCC



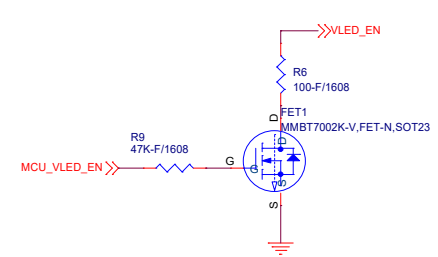
$V_{out} = 2.495(1 + (R2/R1))$, $R1 = RA||RB$
 $5.5V = 2.495(1 + (2/1.65))$, $1.65 = 2 * 6.8 / (2 + 9.5)$

VREF FOR ADC

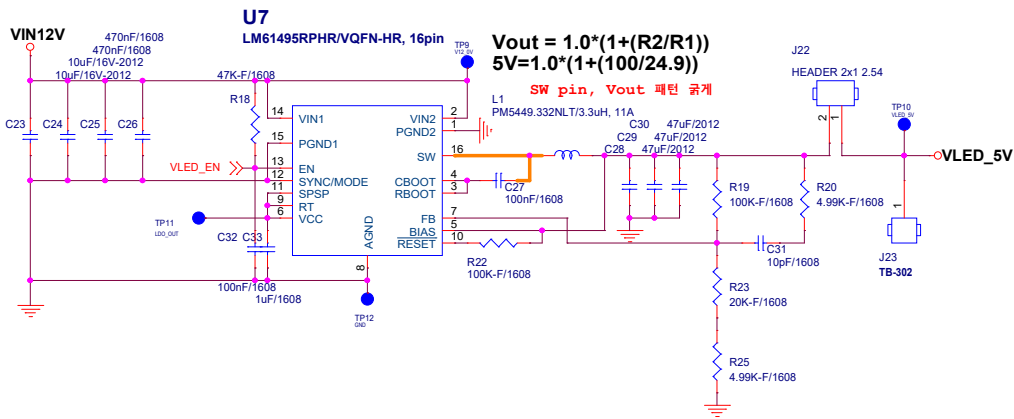


VREF 전원
REF02HSZ 0.5%
REF02CSZ 1%

MCU VLED EN



VLED 5V, Vin 12V, 10A Buck Converter



$V_{out} = 1.0 * (1 + (R2/R1))$
 $5V = 1.0 * (1 + (100/24.9))$

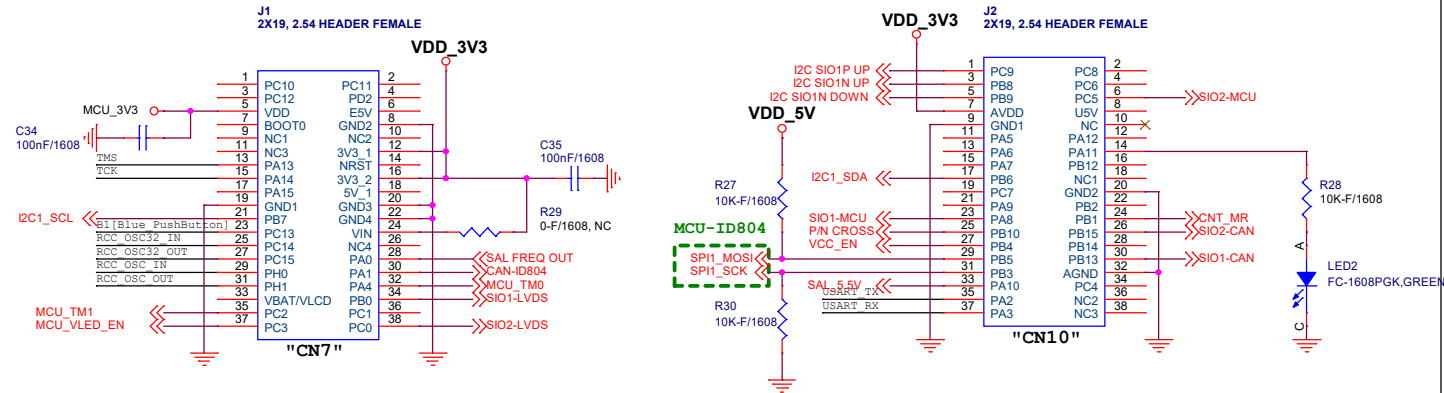
SW pin, Vout 패턴 굵게

Global Technologies
주식회사 글로벌테크놀로지

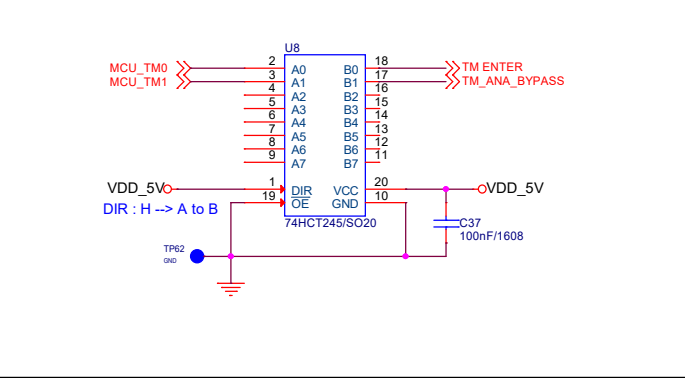
Size	Project name ID804 TEST BD	Rev
Custom	Page name P1. POWER LOGIC	2.0
Sheet	2 of 5	
Date:	Thursday, July 02, 2026	

Design	Check
P.C.G	-

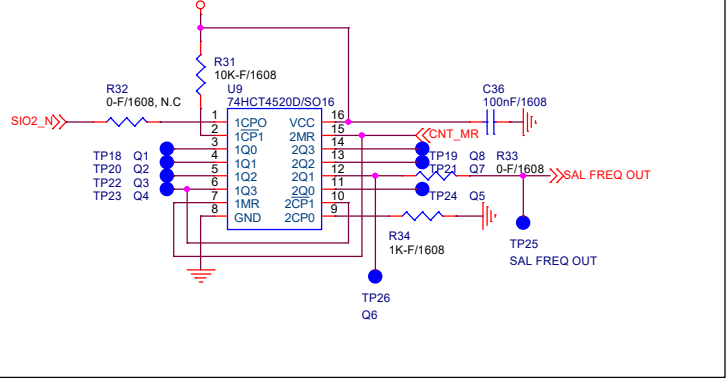
MCU = NUCLEO-F446RE



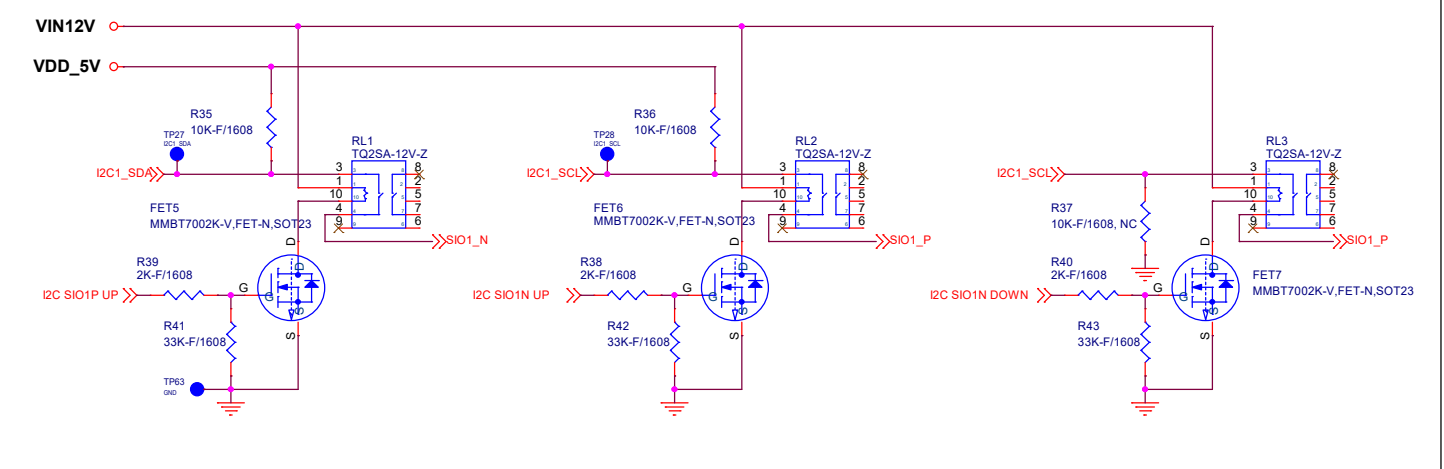
BUFFER - 5V



FREQUENCY DIVIDER_OSC CTL



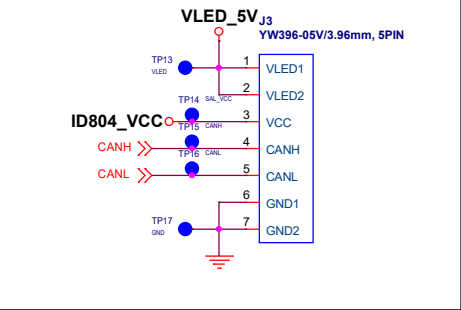
I2C - SIO1 - MCU PIN



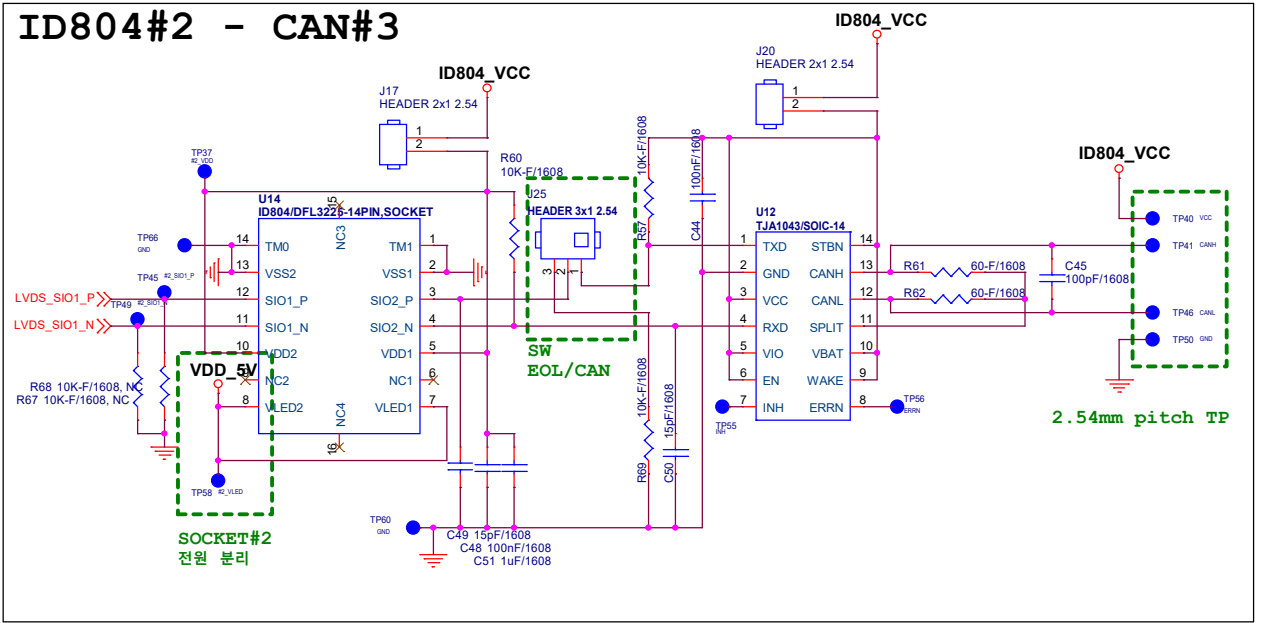
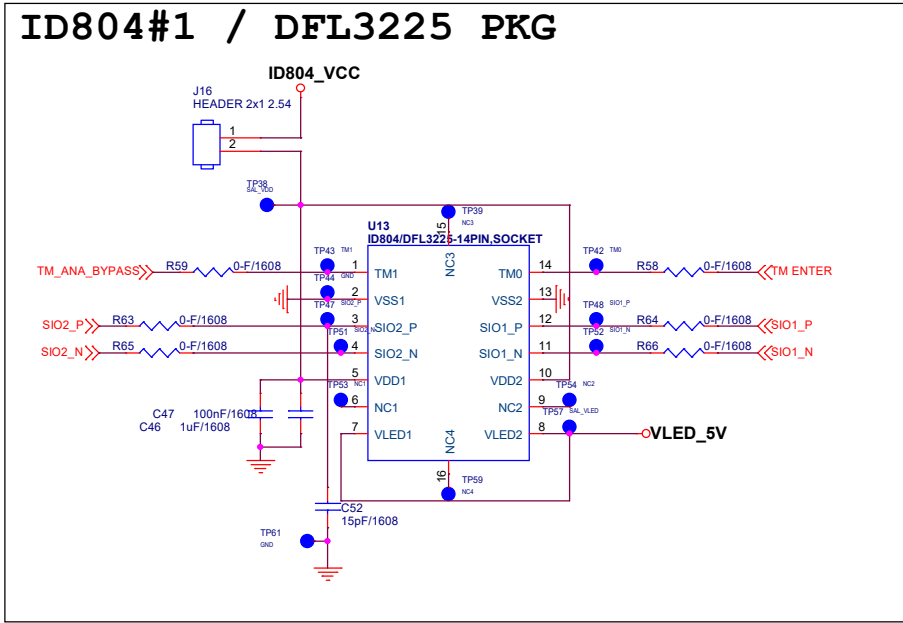
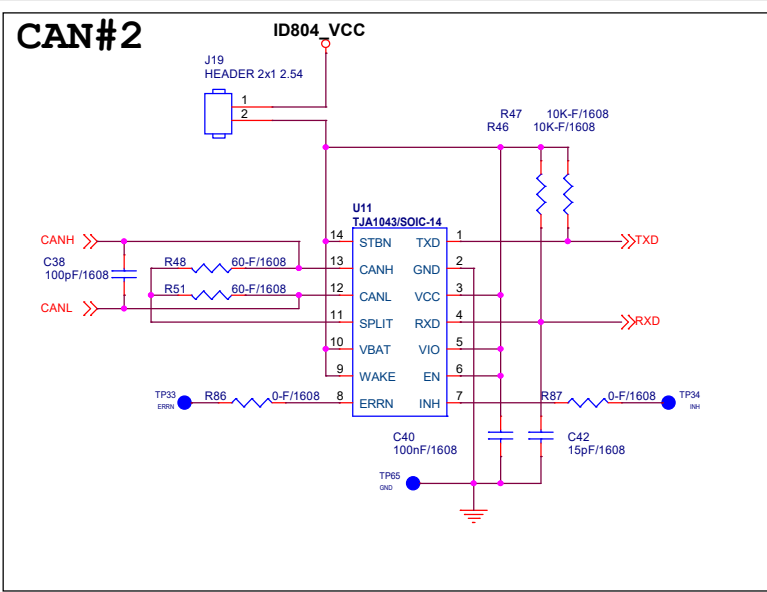
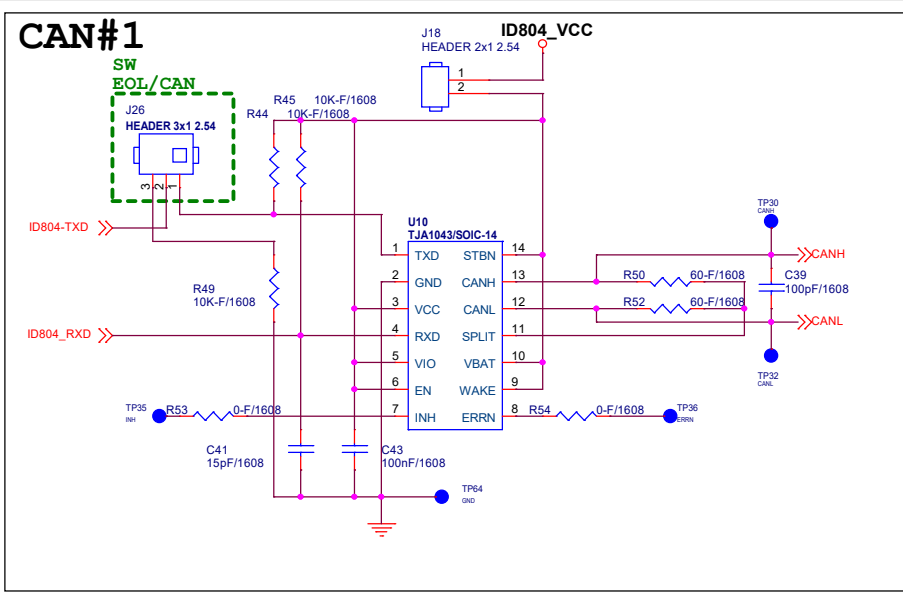
통신모드

PORT	정방향 CAN	역방향 CAN	정방향 LVDS	역방향 LVDS
SIO1-MCU [PA8]	H	L	H	L
SIO2-MCU [PC5]	L	H	L	H
SIO1-CAN [PB13]	L	H	L	L
SIO2-CAN [PB15]	H	L	L	L
SIO1-LVDS [PB0]	L	L	L	H
SIO2-LVDS [PC0]	L	L	H	L
CAN-ID804 [PA1]	H	H	L	L

Connector_CAN



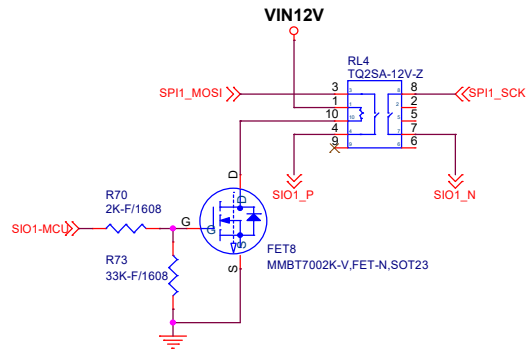
Global Technologies 주요회사 글로벌 테크놀로지		Design	Check
Size A3	Project name ID804 TEST BD	Rev	P.C.G -
	Page name P2. MCU, I2C	2.0	
	Sheet 3 of 5		
	Date Thursday, July 02, 2026		



Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size A3	Project name ID804 TEST BD	Rev	P.C.G	-
	Page name P3. ID804, CAN	2.0		
	Sheet 4 of 5			
	Date: Monday, July 06, 2026			

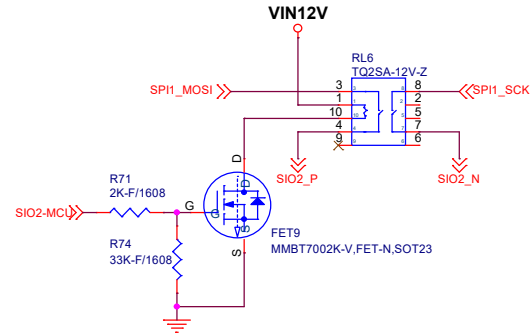
MCU 정방향

MCU - ID804#1_SIO1



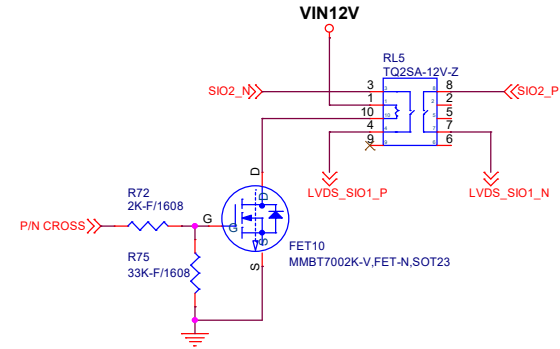
MCU 역방향

MCU - ID804#1_SIO2



P/N CROSS

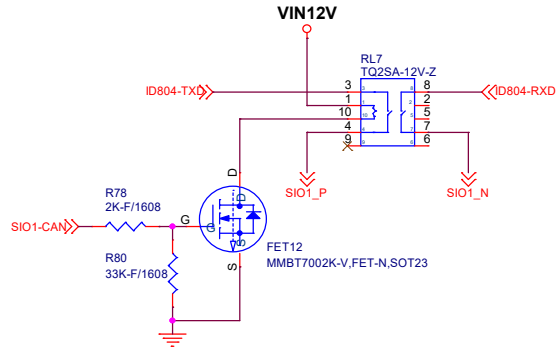
ID804#1_SIO2 - ID804#2



P/N CROSS RELAY 추가

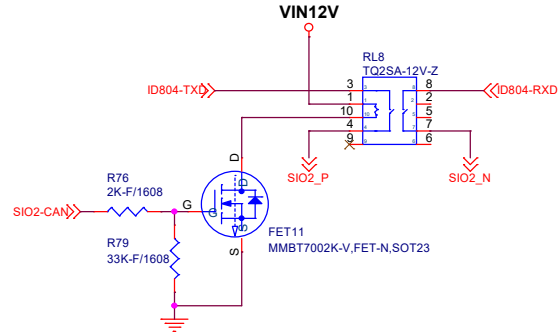
CAN 역방향

ID804#1_SIO1 - CAN#1



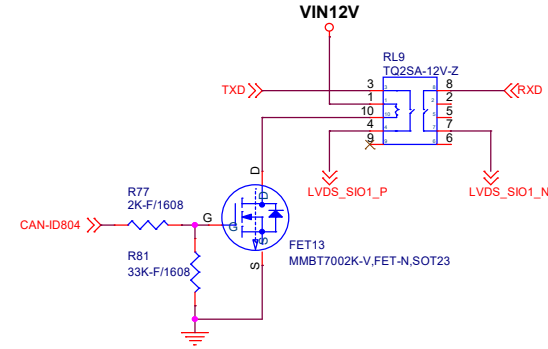
CAN 정방향

ID804#1_SIO2 - CAN#1



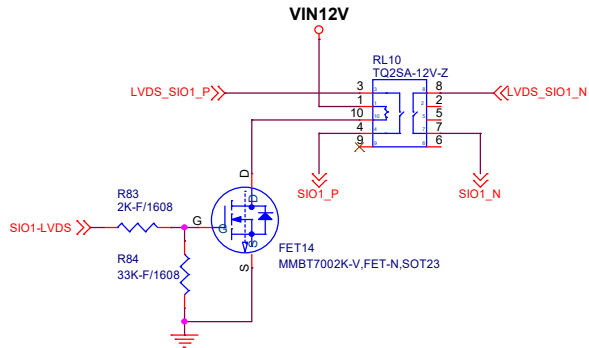
CAN TEST

CAN#2 - ID804#2



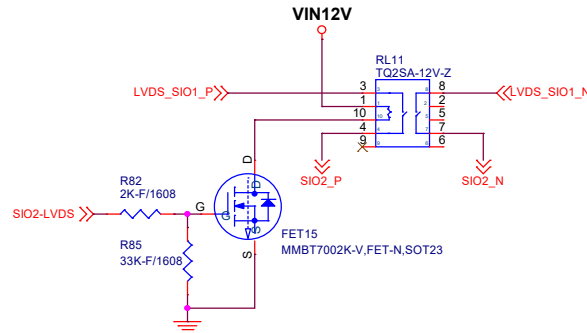
LVDS 역방향

ID804#1_SIO1 - ID804#2



LVDS 정방향

ID804#1_SIO2 - ID804#2



Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size A3	Project name ID804 TEST BD	Rev	P.C.G	-
	Page name P4. RELAY CTRL	2.0		
	Sheet 5 of 5			
	Date Tuesday, July 07, 2026			